

3.1 Working with vectors

- A vector is a quantity which has both magnitude and direction.

These are examples of **vector** quantities:

Quantity	Description	Unit
Displacement	distance in a particular direction	metre (m)
Velocity	rate of change of displacement	metres per second (m s^{-1})
Acceleration	rate of change of velocity	metres per second per second (m s^{-2})
Force/weight	described by magnitude, direction and point of application	newton (N)

- A scalar quantity has magnitude only.

These are examples of **scalar** quantities:

Quantity	Description	Unit
Distance	measure of length	metre (m)
Speed	measure of how quickly a body moves	metres per second (m s^{-1})
Time	measure of ongoing events taking place	second (s)
Mass	measure of the quantity of matter contained in an object	kilogram (kg)

Scalar quantities are always **positive**. When considering motion in a straight line (1-dimensional), **vector** quantities can be **positive** or **negative**.

Example 1

1

SKILLS

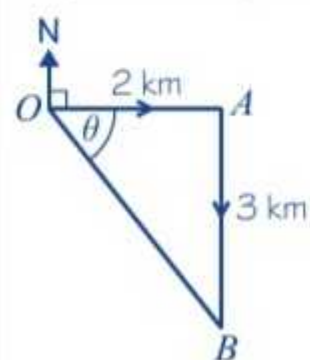
PROBLEM-SOLVING

A girl walks 2 km due east from a fixed point O to A , and then 3 km due south from A to B . Find the total distance walked and describe the displacement of B from O .

The distance the girl has walked is

$$2 \text{ km} + 3 \text{ km} = 5 \text{ km}$$

Representing the girl's journey on a diagram:



$$OB^2 = OA^2 + AB^2$$

$$OB^2 = 4 + 9 = 13$$

So the distance OB is 3.61 km (3 s.f.)

$$\tan \angle AOB = \frac{3}{2}$$

$$\angle AOB = 56.3^\circ$$

B is 3.61 km from O on a bearing of 146° .

Note that the distance of B from O is not the same distance as the girl has walked.

A diagram is essential and gives a clear representation of the displacement OB .

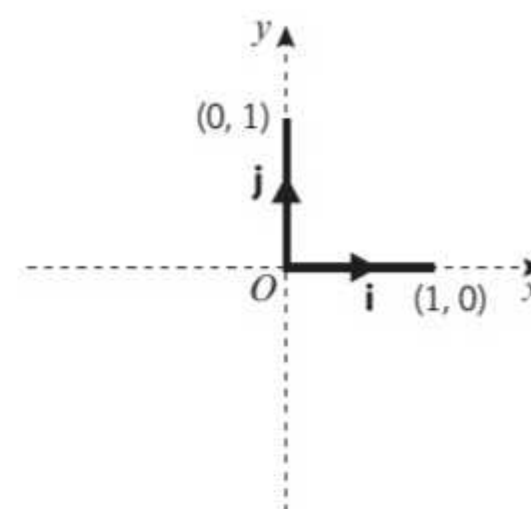
Using Pythagoras' theorem

If the question does not ask you to specify the direction in a particular form, then any method is acceptable.

A three figure bearing is always measured clockwise from north.

You can describe vectors using **i** and **j** notation.

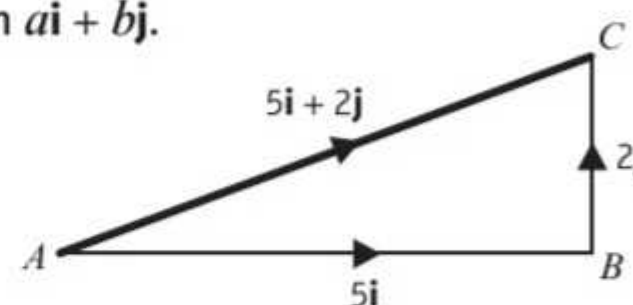
- A unit vector is a vector of length 1. The unit vectors along the Cartesian axes are usually denoted by **i** and **j** respectively.



- You can write any two-dimensional vector in the form $a\mathbf{i} + b\mathbf{j}$.

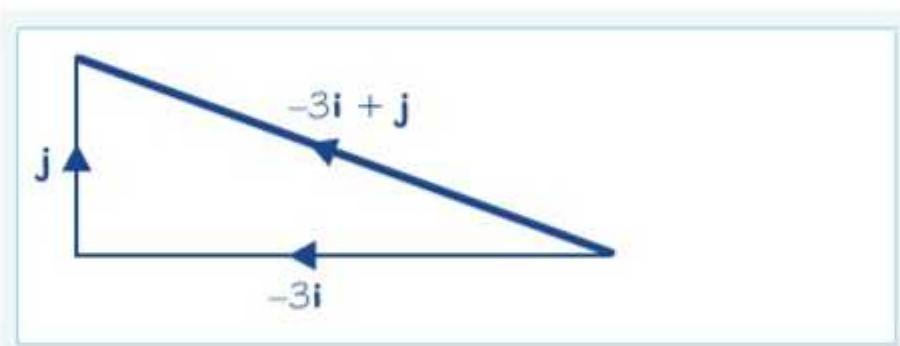
By the triangle law of addition:

$$\begin{aligned}\vec{AC} &= \vec{AB} + \vec{BC} \\ &= 5\mathbf{i} + 2\mathbf{j}\end{aligned}$$



Example 2

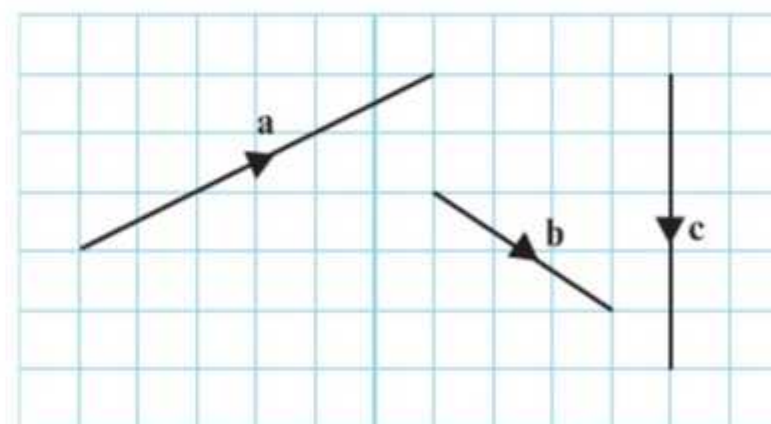
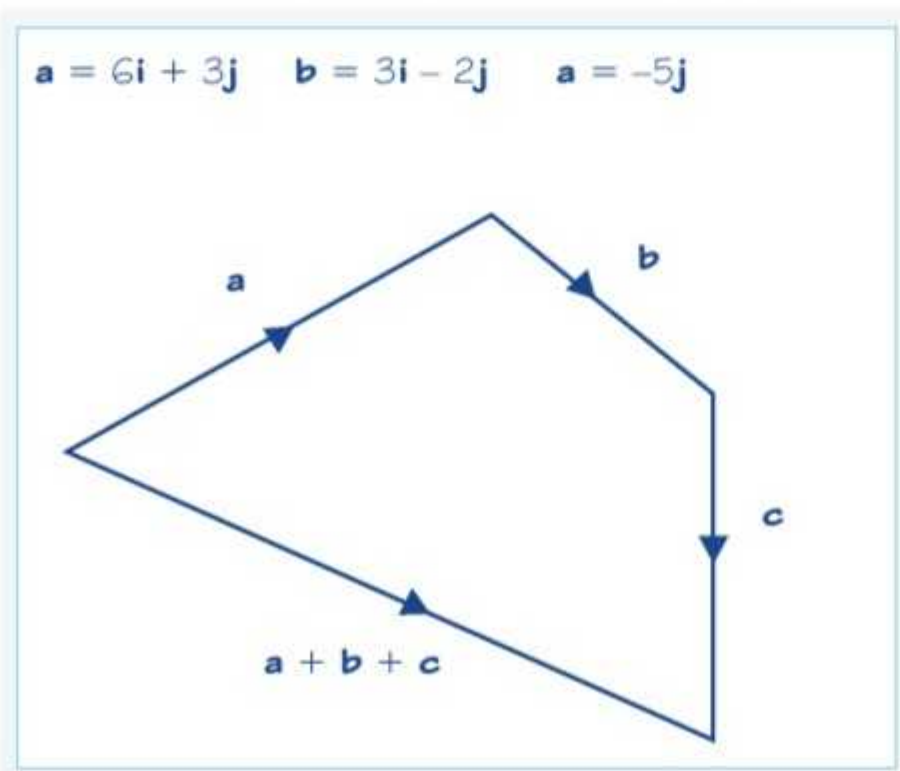
Draw a diagram to represent the vector $-3\mathbf{i} + \mathbf{j}$.



3 units in the direction of the unit vector $-\mathbf{i}$ and 1 unit in the direction of the unit vector $\mathbf{j}$.

Example 3

The diagram shows vectors **a**, **b** and **c**. Draw a diagram to illustrate the vector addition $\mathbf{a} + \mathbf{b} + \mathbf{c}$.



Add the vectors together and collect up **i** and **j** components.

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = 6\mathbf{i} + 3\mathbf{j} + 3\mathbf{i} - 2\mathbf{j} - 5\mathbf{j} = 9\mathbf{i} - 4\mathbf{j}$$

Exercise

3A

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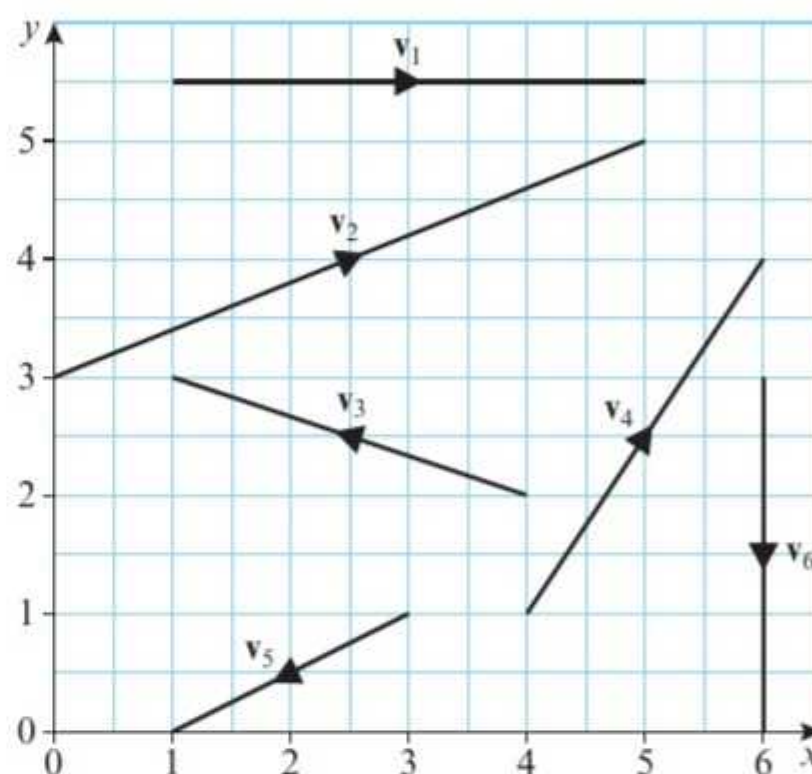
PROBLEM-SOLVING

- 1 A bird flies 5 km due north and then 7 km due east. How far is the bird from its original position, and in what direction?
- 2 A girl cycles 4 km due west and then 6 km due north. Calculate the total distance she has cycled, and her displacement from her starting point.

- 3 a Express the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$, $\mathbf{v}_5$ and $\mathbf{v}_6$ using the $\mathbf{i}$ and $\mathbf{j}$ notation.

b Find:

- i $\mathbf{v}_1 + \mathbf{v}_2$
- ii $\mathbf{v}_4 + \mathbf{v}_5$
- iii $\mathbf{v}_6 + \mathbf{v}_1 + \mathbf{v}_5$



- 4 Find the magnitude of each of these vectors:

- | | | | |
|-------------------------------|-------------------------------|--------------------------------|-------------------------------|
| a $3\mathbf{i} + 4\mathbf{j}$ | b $6\mathbf{i} - 8\mathbf{j}$ | c $5\mathbf{i} + 12\mathbf{j}$ | d $2\mathbf{i} + 4\mathbf{j}$ |
| e $3\mathbf{i} - 5\mathbf{j}$ | f $4\mathbf{i} + 7\mathbf{j}$ | g $-3\mathbf{i} + 5\mathbf{j}$ | h $-4\mathbf{i} - \mathbf{j}$ |

- 5 $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$, $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j}$ and $\mathbf{c} = 5\mathbf{i} - \mathbf{j}$. Find the exact value of the magnitude of:

- | | | |
|-----------------------------|------------------------------|-------------------------------|
| a $\mathbf{a} + \mathbf{b}$ | b $2\mathbf{a} - \mathbf{c}$ | c $3\mathbf{b} - 2\mathbf{c}$ |
|-----------------------------|------------------------------|-------------------------------|

- 6 Find the angle that each of these vectors makes with the positive x -axis:

- | | | | |
|-------------------------------|-------------------------------|--------------------------------|-------------------------------|
| a $3\mathbf{i} + 4\mathbf{j}$ | b $6\mathbf{i} - 8\mathbf{j}$ | c $5\mathbf{i} + 12\mathbf{j}$ | d $2\mathbf{i} + 4\mathbf{j}$ |
|-------------------------------|-------------------------------|--------------------------------|-------------------------------|

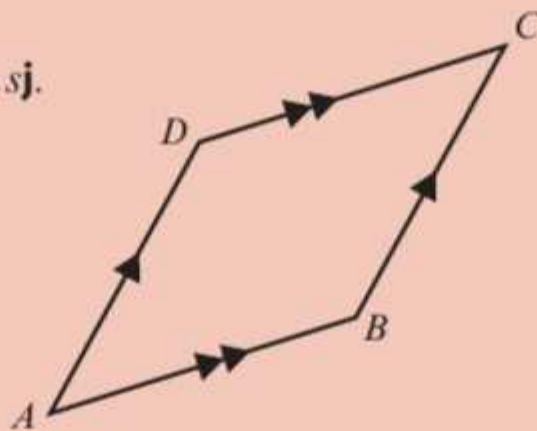
- 7 Find the angle that each of these vectors makes with the positive y -axis:

- | | | | |
|-------------------------------|-------------------------------|--------------------------------|-------------------------------|
| a $3\mathbf{i} - 5\mathbf{j}$ | b $4\mathbf{i} + 7\mathbf{j}$ | c $-3\mathbf{i} + 5\mathbf{j}$ | d $-4\mathbf{i} - \mathbf{j}$ |
|-------------------------------|-------------------------------|--------------------------------|-------------------------------|

Challenge

In the diagram, $\overrightarrow{AB} = p\mathbf{i} + q\mathbf{j}$ and $\overrightarrow{AD} = r\mathbf{i} + s\mathbf{j}$. $ABCD$ is a parallelogram.

Prove that the area of $ABCD$ is $ps - qr$.



Problem-solving

Draw the parallelogram on a coordinate grid, and choose a position for the origin that will simplify your calculations.

3.2 Solving problems with vectors written using **i** and **j** notation

- When vectors are written in terms of the unit vectors **i** and **j** you can add them together by adding the terms in **i** and **j** separately. You can subtract vectors in a similar way.

Example 4

Given $\mathbf{p} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{q} = 5\mathbf{i} + \mathbf{j}$, find $\mathbf{p} + \mathbf{q}$ in terms of **i** and **j**.

$$\begin{aligned}\mathbf{p} + \mathbf{q} &= (2\mathbf{i} + 3\mathbf{j}) + (5\mathbf{i} + \mathbf{j}) \\ &= (2\mathbf{i} + 5\mathbf{i}) + (3\mathbf{j} + \mathbf{j}) \\ &= 7\mathbf{i} + 4\mathbf{j}\end{aligned}$$

Rearrange with the **i** terms together and the **j** terms together.

Simplify the answer.

Example 5

Given $\mathbf{a} = 5\mathbf{i} + 2\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j}$, find $2\mathbf{a} - \mathbf{b}$ in terms of **i** and **j**.

$$\begin{aligned}2\mathbf{a} &= 2(5\mathbf{i} + 2\mathbf{j}) \\ &= 10\mathbf{i} + 4\mathbf{j} \\ 2\mathbf{a} - \mathbf{b} &= (10\mathbf{i} + 4\mathbf{j}) - (3\mathbf{i} - 4\mathbf{j}) \\ &= (10\mathbf{i} - 3\mathbf{i}) + (4\mathbf{j} - (-4\mathbf{j})) \\ &= 7\mathbf{i} + 8\mathbf{j}\end{aligned}$$

When you double **a**, the **i** term and the **j** term are both doubled – just the same as when you multiply out a bracket in algebra.

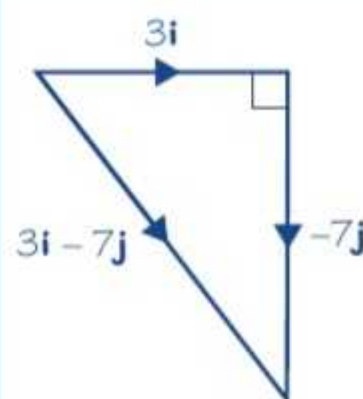
Rearrange with the **i** terms together and the **j** terms together.

Simplify the answer but take care when subtracting a negative term.

- When a vector is written in terms of the unit vectors **i** and **j** you can find the magnitude using Pythagoras' Theorem. The magnitude of vector **a** is written $|\mathbf{a}|$.

Example 6

Given $\mathbf{v} = 3\mathbf{i} - 7\mathbf{j}$, find the magnitude of **v**.



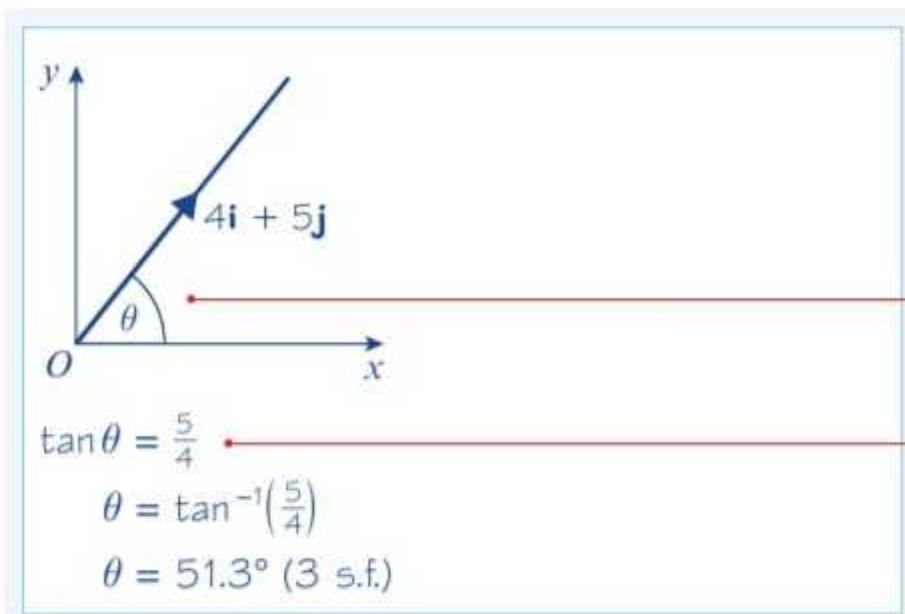
$$\begin{aligned}|\mathbf{v}| &= \sqrt{3^2 + (-7)^2} \\ &= \sqrt{58} \\ &= 7.62 \text{ (3 s.f.)}\end{aligned}$$

Because **i** and **j** are perpendicular vectors, you have a right-angled triangle.

$|\mathbf{v}|$ means the magnitude of **v**.

Example 7

Find the angle between the vector $4\mathbf{i} + 5\mathbf{j}$ and the positive x -axis.



Hint Always draw a diagram first.

Identify the angle you need to find.

We have a right-angled triangle, so finding the angle is straightforward.

Example 8

Given that $\mathbf{p} = 4\mathbf{i} + 3\mathbf{j}$ and $\mathbf{q} = 2\mathbf{i} - 6\mathbf{j}$, find λ if $\mathbf{p} + \lambda\mathbf{q}$ is parallel to the vector:

a $\mathbf{j}$ **b** $\mathbf{i} + \mathbf{j}$

a $\mathbf{p} = 4\mathbf{i} + 3\mathbf{j}, \mathbf{q} = 2\mathbf{i} - 6\mathbf{j}$
 $\mathbf{p} + \lambda\mathbf{q} = 4\mathbf{i} + 3\mathbf{j} + \lambda(2\mathbf{i} - 6\mathbf{j})$
 $\quad = 4\mathbf{i} + 2\lambda\mathbf{i} + 3\mathbf{j} - 6\lambda\mathbf{j}$
 $4\mathbf{i} + 2\lambda\mathbf{i} = \mathbf{0}$
 $2\lambda\mathbf{i} = -4\mathbf{i}$
 $\Rightarrow \lambda = -2$

b $\mathbf{p} = 4\mathbf{i} + 3\mathbf{j}$
 $\mathbf{q} = 2\mathbf{i} - 6\mathbf{j} \Rightarrow \lambda\mathbf{q} = 2\lambda\mathbf{i} - 6\lambda\mathbf{j}$
 $\mathbf{p} + \lambda\mathbf{q} = (4\mathbf{i} + 3\mathbf{j}) + (2\lambda\mathbf{i} - 6\lambda\mathbf{j})$
 $\quad = \mathbf{i}(4 + 2\lambda) + \mathbf{j}(3 - 6\lambda)$
 $4 + 2\lambda = 3 - 6\lambda$
 $\Rightarrow \lambda = -\frac{1}{8}$

Add together $\mathbf{p}$ and $\lambda\mathbf{q}$.

$\mathbf{p} + \lambda\mathbf{q}$ is parallel to $\mathbf{j}$, therefore the $\mathbf{i}$ -component of $\mathbf{p} + \lambda\mathbf{q}$ is 0.

We require the vector in the form $\mathbf{p} + \lambda\mathbf{q}$ so multiply $\mathbf{q}$ through by λ .

Rearrange with the $\mathbf{i}$ terms together and the $\mathbf{j}$ terms together.

The $\mathbf{i}$ and $\mathbf{j}$ components must be equal so equate them and solve for λ .

Exercise 3B

1 Given that $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 4\mathbf{i} - \mathbf{j}$, find these in terms of $\mathbf{i}$ and $\mathbf{j}$.

a $\mathbf{a} + \mathbf{b}$

b $3\mathbf{a} + \mathbf{b}$

c $2\mathbf{a} - \mathbf{b}$

d $2\mathbf{b} - \mathbf{a}$

e $3\mathbf{a} - 2\mathbf{b}$

f $\mathbf{b} - 3\mathbf{a}$

g $4\mathbf{b} - \mathbf{a}$

h $2\mathbf{a} - 3\mathbf{b}$

2 Given that $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$, find:

a λ if $\mathbf{a} + \lambda\mathbf{b}$ is parallel to the vector $\mathbf{i}$

b μ if $\mu\mathbf{a} + \mathbf{b}$ is parallel to the vector $\mathbf{j}$.

- 3 Given that $\mathbf{c} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{d} = \mathbf{i} - 2\mathbf{j}$, find:
- a λ if $\mathbf{c} + \lambda\mathbf{d}$ is parallel to $\mathbf{i} + \mathbf{j}$ b μ if $\mu\mathbf{c} + \mathbf{d}$ is parallel to $\mathbf{i} + 3\mathbf{j}$
 c s if $\mathbf{c} - s\mathbf{d}$ is parallel to $2\mathbf{i} + \mathbf{j}$ d t if $\mathbf{d} - t\mathbf{c}$ is parallel to $-2\mathbf{i} + 3\mathbf{j}$.
- 4 In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively. Find the magnitude and bearing of these vectors:
- a $2\mathbf{i} + 3\mathbf{j}$ b $4\mathbf{i} - \mathbf{j}$ c $-3\mathbf{i} + 2\mathbf{j}$ d $-2\mathbf{i} - \mathbf{j}$

Challenge

Given that $\mathbf{a} = 3\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = \mathbf{i} + 4\mathbf{j}$, find the values of λ if $\mathbf{a} + \lambda\mathbf{b}$ is perpendicular to $3\mathbf{i} - 2\mathbf{j}$.

Comment on your answer.

SKILLS

ADAPTIVE LEARNING

SKILLS

EXECUTIVE FUNCTION

3.3 The velocity of a particle as a vector

- The velocity of a particle is a vector in the direction of motion. Its magnitude is the speed of the particle. The velocity is usually denoted by $\mathbf{v}$.

If a particle is moving with constant velocity $\mathbf{v} \text{ m s}^{-1}$, then after time t seconds it will have moved $\mathbf{vt} \text{ m}$. The displacement is parallel to the velocity. The magnitude of the displacement is the distance from the starting point.

Example 9

A particle is moving with constant velocity $\mathbf{v} = (3\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$. Find **a** the speed of the particle, **b** the distance moved every 4 seconds.

a $\mathbf{v} = (3\mathbf{i} + \mathbf{j})$,
 so speed $|\mathbf{v}| = \sqrt{3^2 + 1^2} = \sqrt{10}$
 $= 3.16 \text{ m s}^{-1}$ (3 s.f.)

The speed is the magnitude of the velocity.

b Method 1

Displacement $= \mathbf{vt} = (3\mathbf{i} + \mathbf{j}) \times 4$
 $= (12\mathbf{i} + 4\mathbf{j})$

Find the displacement of the particle in 4 seconds.

Therefore, the distance moved is given by

$|\mathbf{vt}| = \sqrt{12^2 + 4^2} = \sqrt{144 + 16} = \sqrt{160}$
 $= 12.6 \text{ m}$ (3 s.f.)

Displacement $= \mathbf{v} \times t$

Distance is magnitude of displacement.

Method 2

$\mathbf{v} = (3\mathbf{i} + \mathbf{j})$,
 so speed $= |\mathbf{v}| = \sqrt{3^2 + 1^2} = \sqrt{10}$
 $= 3.16 \text{ m s}^{-1}$

Find the speed of the particle.

Distance $= 3.16 \times 4 = 12.6 \text{ m}$ (3 s.f.)

Distance = speed $\times$ time

Use the unrounded value in your calculation.

Exercise

3C

SKILLS

EXECUTIVE FUNCTION

- Find the speed of a particle moving with these velocities:
a $(3\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$ **b** $(24\mathbf{i} - 7\mathbf{j})\text{ km h}^{-1}$ **c** $(5\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$ **d** $(-7\mathbf{i} + 4\mathbf{j})\text{ cm s}^{-1}$
- Find the distance moved by a particle which travels for:
a 5 hours at velocity $(8\mathbf{i} + 6\mathbf{j})\text{ km h}^{-1}$ **b** 10 seconds at velocity $(5\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$
c 45 minutes at velocity $(6\mathbf{i} + 2\mathbf{j})\text{ km h}^{-1}$ **d** 2 minutes at velocity $(-4\mathbf{i} - 7\mathbf{j})\text{ cm s}^{-1}$
- Find the speed and the distance travelled by a particle moving with:
a velocity $(-3\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$ for 15 seconds **b** velocity $(2\mathbf{i} + 5\mathbf{j})\text{ m s}^{-1}$ for 3 seconds
c velocity $(5\mathbf{i} - 2\mathbf{j})\text{ km h}^{-1}$ for 3 hours **d** velocity $(12\mathbf{i} - 5\mathbf{j})\text{ km h}^{-1}$ for 30 minutes

3.4 Solving problems involving velocity and time using vectors

- If a particle starts from the point with position vector $\mathbf{r}_0$ and moves with constant velocity $\mathbf{v}$, then its displacement from its initial position at time t is $\mathbf{v}t$ and its position vector $\mathbf{r}$ is given by:

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$$

Example

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SKILLS

PROBLEM-SOLVING

A particle starts from the point with position vector $(3\mathbf{i} + 7\mathbf{j})\text{ m}$ and moves with constant velocity $(2\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$. Find the position vector of the particle 4 seconds later.

$$\begin{aligned}\text{Displacement} &= \mathbf{v}t = 4(2\mathbf{i} - \mathbf{j}) = 8\mathbf{i} - 4\mathbf{j} \\ \text{Position vector } \mathbf{r} &= (3\mathbf{i} + 7\mathbf{j}) + (8\mathbf{i} - 4\mathbf{j}) \\ &= (3 + 8)\mathbf{i} + (7 - 4)\mathbf{j} \\ &= 11\mathbf{i} + 3\mathbf{j}\end{aligned}$$

$$\text{Distance} = \mathbf{v} \times t$$

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$$

Position vector after 4 seconds = position vector of starting point + displacement.

Example

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A particle moving at a constant velocity is at the point with position vector $(2\mathbf{i} - 4\mathbf{j})\text{ m}$ at time $t = 0$. The particle is moving at a constant velocity. Five seconds later it is at the point with position vector $(12\mathbf{i} + 16\mathbf{j})\text{ m}$. Find the velocity of the particle.

$$\begin{aligned}\text{Displacement} &= (12\mathbf{i} + 16\mathbf{j}) - (2\mathbf{i} - 4\mathbf{j}) \\ &= (12 - 2)\mathbf{i} + (16 + 4)\mathbf{j} \\ &= 10\mathbf{i} + 20\mathbf{j}\end{aligned}$$

Travels $10\mathbf{i} + 20\mathbf{j}$ in 5 seconds, so

$$\mathbf{v} = \frac{1}{5}(10\mathbf{i} + 20\mathbf{j}) = (2\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$$

You need to use the same formula $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$, but this time you are using it to find $\mathbf{v}$.

Start by finding the displacement.

Travelling at constant velocity, so divide displacement by time taken to obtain $\mathbf{v}$.

Example 12

At time $t = 0$, a particle has position vector $4\mathbf{i} + 7\mathbf{j}$ and is moving with speed 15 m s^{-1} in the direction $3\mathbf{i} - 4\mathbf{j}$. Find its position vector after 2 seconds.

The magnitude of vector $3\mathbf{i} - 4\mathbf{j}$ is
 $\sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5$.
 So a unit vector in the direction of motion is
 $\frac{1}{5}(3\mathbf{i} - 4\mathbf{j})$, and the velocity is
 $15 \times \frac{1}{5}(3\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1} = (9\mathbf{i} - 12\mathbf{j}) \text{ m s}^{-1}$.
 New position $= (4\mathbf{i} + 7\mathbf{j}) + 2(9\mathbf{i} - 12\mathbf{j})$
 $= (4\mathbf{i} + 7\mathbf{j}) + (18\mathbf{i} - 24\mathbf{j})$
 $= (4 + 18)\mathbf{i} + (7 - 24)\mathbf{j}$
 $= 22\mathbf{i} - 17\mathbf{j}$

Start by using the speed and the direction of motion to find the velocity.

You need to start with a unit vector in the direction of motion, then multiply by the speed. This gives you a vector in the right direction with the required magnitude.

Now use $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

- The acceleration of a particle tells you how the velocity changes with time. Acceleration is a vector, usually denoted by $\mathbf{a}$. If a particle with initial velocity $\mathbf{u}$ moves with constant acceleration $\mathbf{a}$ then its velocity, $\mathbf{v}$ at time t is given by:

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

Example 13

A particle P has velocity $(-3\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$ at time $t = 0$. The particle moves with constant acceleration $\mathbf{a} = (2\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-2}$. Find the speed of the particle after 3 seconds.

After 3 seconds, the velocity of P is
 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$
 $= (-3\mathbf{i} + \mathbf{j}) + 3 \times (2\mathbf{i} + 3\mathbf{j})$
 $= (-3\mathbf{i} + \mathbf{j}) + (6\mathbf{i} + 9\mathbf{j})$
 $= (3\mathbf{i} + 10\mathbf{j}) \text{ m s}^{-1}$
 So the speed of $P = \sqrt{3^2 + 10^2} = \sqrt{109}$
 $= 10.4 \text{ m s}^{-1}$ to 3 s.f.

Using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

The speed of P is the magnitude of its velocity.

- A force applied to a particle has both magnitude and direction, so force is also a vector. The force causes the particle to accelerate:

$\mathbf{F} = m\mathbf{a}$, where m is the mass of the particle.

Example 14

A constant force, $\mathbf{F} \text{ N}$, acts on a particle of mass 2 kg for 10 seconds. The particle is initially at rest, and 10 seconds later it has velocity $(10\mathbf{i} - 24\mathbf{j}) \text{ m s}^{-1}$. Find $\mathbf{F}$.

$(10\mathbf{i} - 24\mathbf{j}) = 10\mathbf{a}$
 So $\mathbf{a} = (\mathbf{i} - 2.4\mathbf{j}) \text{ m s}^{-2}$
 So $\mathbf{F} = 2 \times (\mathbf{i} - 2.4\mathbf{j}) = (2\mathbf{i} - 4.8\mathbf{j}) \text{ N}$

Use $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ to find the acceleration of the particle. $\mathbf{u} = 0$ as the particle is initially at rest.

Using $\mathbf{F} = m\mathbf{a}$